



Magical Trees & The 3 Travelers



1. Traversal Alag Kyu Hai? 🤔

Arrays ya Linked List ek seedhi sadak (Linear) ki tarah hote hain - bas straight chalo! Par **Tree** ek non-linear maze (bhool-bhulaiya) hai jisme bohot saare raaste (children) hote hain.

UNIVERSAL RULE:

Left bache (subtree) se hamesha Right bache se **PEHLE** milna hai!

2. Naam Mein Hi Magic Hai! ✨ (The 3 Ways)

Traversal ka naam is baat par rakha gaya hai ki humare **Captain ROOT** ki entry kab hoti hai:

🥇 Pre-Order (Root Sabse Pehle)

Order: **ROOT** → Left → Right

🥈 In-Order (Root Beech Mein)

Order: Left → **ROOT** → Right

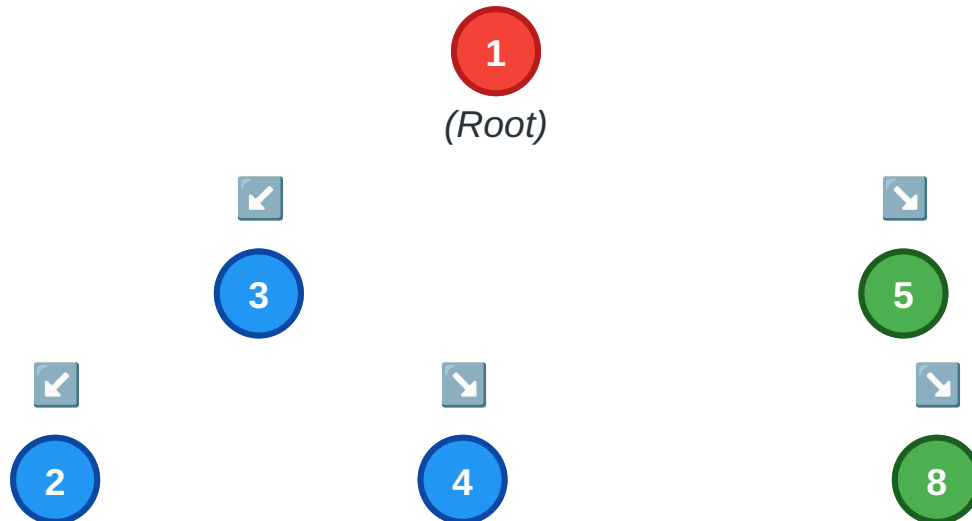
🥉 Post-Order (Root Sabse Aakhiri)

Order: Left → Right → **ROOT**

Jadoo (Use Case): Jab tree ko Delete (memory free) karna ho, ya bottom-up calculation karni ho.

Homemade Tree Diagram

Ek simple tree aur uske teeno traversal ke results dekhte hain:



Preorder : 1, 3, 2, 4, 5, 8

Inorder : 2, 3, 4, 1, 5, 8

Postorder: 2, 4, 3, 8, 5, 1

The Grand Memory Link

Dono Concepts (BST + Traversals) Ek Saath Kaise Jude Hain?

Socho tumhare paas ek **Binary Search Tree (BST)** hai jisme *Left chota aur Right bada* hota hai.

Agar tum uspe **INORDER TRAVERSAL (Left → Root → Right)** ka magic spell cast karoge, toh output hamesha ek **Sorted (Ascending order)** array nikal ke aayega!

Yehi interview ka sabse bada secret hai! 🤔